Route Wise — Market Requirements Document (MRD)

DOCUMENT: MRD-v2.0
TEAM: Cyber Knight
AUDIENCE: Mentors & Judges
STATUS: Final Pitch Ready

The Business Case, Market Need & Real-World Problem — Explaining Route Wise in Simple English

The 30-Second Elevator Pitch for Judges: Most navigation apps blindly suggest the shortest route, even if it leads straight into heavy junction gridlock. **Route Wise (Team Cyber Knight)** is an intelligent traffic advisory platform and live dashboard that evaluates multiple driving alternatives, penalizes severe junction bottlenecks, and predicts 15-minute congestion trends using an official public **ARIMA (1,0,1)** ML model from GitHub. It explains *why* it chose the route in plain words (e.g. *'Saves 14 mins travel time and 37 km shorter vs Route #2'*), saving time, fuel, and driver frustration for daily commuters and couriers.

1. Real-World Story: Why Traditional GPS Fails Drivers

The Scenario (Meet Karthik the Daily Commuter & Courier): It is 5:15 PM peak evening rush hour in Tiruchirappalli. Karthik has an urgent trip / package delivery from Central Bus Stand to Cantonment & Thillai Nagar.

• **What Traditional GPS Does:** Recommends Route A (Direct City Arterial) because it is only **10.0 km**. However, City Arterial has an unexpected junction jam adding **18 minutes** of bumper-to-bumper delay. Total travel time: **42 minutes**. Karthik arrives late, fuel is burned idling, and stress peaks.

• **What Route Wise Does:** Evaluates 3 alternatives simultaneously. Recognizes Route B (Outer Ring Corridor) is **12.5 km** (2.5 km longer) but has near-zero delay (only **2 mins**). Total travel time: **26 minutes**. Route Wise recommends Route B, explaining: *'Fastest path — Avoids 16 minutes of junction gridlock'*. Karthik arrives 16 minutes early!

Navigation Strategy	Distance	Traffic Delay	Total Time	Driver & Commuter Outcome
Standard Navigation App (Picks shortest distance)	10.0 km (Shortest)	**+18 mins** (Heavy gridlock)	**42 mins**	Late arrival, high stress, fuel wasted idling in junction bottleneck.
Route Wise (Cyber Knight) (Multi-criteria AI + ARIMA)	12.5 km (+2.5 km)	**+2 mins** (Smooth flowing)	**26 mins (16m faster!)**	On-time arrival, clear trade-off explanation, fuel-efficient constant speed.

2. The Big Market Problem We Solve

1. Opaque 'Black-Box' Algorithms: Traditional apps give a single route without explaining why it was picked, leaving commuters uncertain.
2. Inability to Predict Traffic Spikes: Existing maps show current delay, but cannot tell if congestion will clear or worsen in 15 minutes.
3. Demo & Network Fragility: Standard mapping APIs crash when offline or over quota. Developers need high-reliability fail-safes.

3. Market Size: Multi-Billion Dollar Opportunity

TAM: $14.2 Billion Global GIS, Navigation & Fleet Logistics API Market	**SAM: $2.8 Billion** Urban Last-Mile Delivery & Commuter Routing Software	**SOM: $350 Million** Tier-2 & Tier-1 Indian Logistics SaaS, Startups & Commuters

4. Target Audience, Competition & Revenue Model

4.1 Who Needs This? (Target Customer Segments)

- **Daily Urban Commuters (2W & 4W):** Need to bypass known junction bottlenecks during 8:00 AM – 10:00 AM and 5:00 PM – 7:30 PM peak commutes.
- **On-Demand Couriers & Delivery Drivers:** Under strict SLAs (8 deliveries/hr). A 15-minute junction delay causes missed order deadlines and penalties.
- **Fleet & Logistics Dispatchers:** Managing delivery vans. A 15-minute delay per driver compounds into thousands of wasted fuel rupees weekly.

4.2 Why Route Wise Wins Against Big Tech APIs

Feature / Capability	Google Maps API	Mapbox Navigation	Route Wise (Cyber Knight)
Configurable Scoring Weights	No (Fixed default algorithm)	Partial (Limited vehicle profiles)	**Yes — Customize Time, Delay Penalty & Distance**
Transparent 'Why' Rationale	No (Opaque black-box route)	No (No explanation given)	**Yes — Clear tags like 'Saves 14m & 37 km shorter'**
15-Min ML Traffic Forecast	No (Live conditions only)	No (Historical averages only)	**Yes — Public ARIMA(1,0,1) model for +15m, +30m, +45m**
Traffic Severity Badges	Partial (Visual map lines only)	Partial (Generic speed colors)	**Yes — Green / Yellow / Red delay classifier**
Offline Demo Fail-Safe	No (Fails if internet drops)	No (Requires paid offline packs)	**Yes — High-fidelity local coordinate dictionary**
Side-by-Side Comparison Matrix	No (Must switch back & forth)	No (Single route view)	**Yes — Simultaneous comparison table**

4.3 How We Monetize (Simple Tiered Model)

Tier	Price	Included Quota & Capabilities	Target Customer
Free Developer	$0 / mo	1,000 queries/month, 60 req/min rate limit, full dashboard access, offline mock mode.	Hackathons, student projects, prototyping.
Growth Fleet	$0.003 / query	Up to 250,000 queries/month, custom scoring presets, 60-second live refresh API, ARIMA forecast.	Courier startups, local delivery fleets.
Enterprise	Custom SLA	Unlimited queries, multi-stop TSP route optimization, dedicated container hosting, 99.9% uptime SLA.	Logistics corporations, mobility providers.

4.4 Top 3 Takeaways to Explain to Judges & Mentors

1. Not Just Another Map: We don't reinvent map tiles; we solve the intelligence layer that evaluates multiple alternatives and tells you *why* Route #1 is the best.	**2. Tangible Commuter ROI:** Saving 15 minutes of junction delay per trip directly reduces fuel consumption, vehicle wear, late penalties, and driver burnout.	**3. Production-Ready Tech:** Powered by TomTom live traffic, public **ARIMA (1,0,1)** ML forecasting, resilient fail-safes, and **40 automated tests** passed at 100%.